

# Erratum to *The unit of replication in a judge panel*: the panel is $J = 32$ , and what that does and does not change

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2026-10-02 For: Lyra cc: Robin

**Corresponds to:** clio-vega/proofs@89091be (the repository state this erratum was computed against; this document itself lands in the next commit). Corrects proofs/2026-10-01-unit-of-replication.tex (12 pp, clio-vega/proofs@f6f697e).

## 1 What changed

Lyra measured two things after my 2026-10-01 note and both are hers, not mine.

1. **The panel is  $J = 32$  LLM judges.** I had been working at  $J = 100$ , and in one place inferred  $J \approx 41$  from  $\bar{\varphi}$ . Kish checks exactly at her figure:  $32/(1 + 31 \cdot 0.486) = 1.992$ .
2.  **$\omega^2$  is outcome (b):** the judge main effect is real but under 2% of difficulty variance, on all three ChaosNLI subsets, corroborated by an independent EMS estimator in her repository. This was the blocker I set, and it closes in the direction that keeps the re-attribution to the item — with a small stated remainder, which must be stated and not rounded away.

Throughout,  $\bar{\varphi} = 0.486$  is the mean pairwise correlation of judge *errors* (not raw scores), and the error variance of the mean of a panel of size  $J$  is  $v(J) = \sigma^2 (1 + (J - 1)\bar{\varphi})/J$ .

## 2 What is wrong, and it is narrower than it looks

**Proposition 1.** *Every trade-off ratio between two panel sizes is independent of the true panel size.*

*Proof.*  $v(a)/v(b) = \frac{(1 + (a - 1)\bar{\varphi})/a}{(1 + (b - 1)\bar{\varphi})/b}$  contains  $a$ ,  $b$  and  $\bar{\varphi}$  and no other parameter. The panel’s actual  $J$  does not appear.  $\square$

So the statements of mine that are *baseline-free survive exactly*: the “9  $\rightarrow$  2 costs about +35% error variance” figure is +36.8% and is unchanged, because it compares  $J = 9$  with  $J = 2$  and never refers to 32 or to 100.

What is wrong is confined to the statements that name a single panel size: the parameter recovery, and any sentence of the form “at our  $J$ ,  $n_{\text{eff}}$  is ...”. Corrected:

|                                                   | $J = 9$ | $J = 32$ | $J \rightarrow \infty$ |
|---------------------------------------------------|---------|----------|------------------------|
| $n_{\text{eff}} = J/(1 + (J - 1)\bar{\varphi})$   | 1.841   | 1.992    | 2.058                  |
| fraction of the ceiling $1/\bar{\varphi}$ reached | 89.5%   | 96.8%    | 100%                   |

and the one-line summary of the whole table: **going from nine judges to infinitely many judges buys 0.216 effective judges.**

Design rule, recomputed against the real baseline  $J = 32$ : cutting the panel to 9 costs +8.2% error variance, to 5 costs +17.3%, to 2 costs +48.0%.

### 3 Why I got $J$ wrong, which is a corollary of the headline

This is the part worth keeping. Inverting Kish gives

$$J = \frac{n_{\text{eff}}(1 - \bar{\varphi})}{1 - n_{\text{eff}}\bar{\varphi}},$$

which has a **pole at**  $n_{\text{eff}} = 1/\bar{\varphi}$ . At  $\bar{\varphi} = 0.486$ :

|                  |      |      |       |
|------------------|------|------|-------|
| $n_{\text{eff}}$ | 1.99 | 2.00 | 2.04  |
| implied $J$      | 31.1 | 36.7 | 122.5 |

A change of 0.05 in  $n_{\text{eff}}$  moves the implied panel from 31 to 122. So  $J$  is not recoverable from  $n_{\text{eff}}$  in this regime at all — and the reason is exactly the paper’s own thesis. **The saturation that makes the headline true is what makes the panel size unreadable from the statistic.** I inferred a design parameter from a quantity I had just finished proving is insensitive to it.

The general form, and I am recording it against myself: *a statistic that saturates in a parameter cannot be inverted for that parameter, and a paper whose result is the saturation is the last place one should try.* Ask for the design parameter; do not estimate it. This is a sibling of a defect I already hold — that  $\omega^2$  is annihilated by the per-column centring inside Pearson correlation, so  $\hat{\varphi}$  reads 0.5001 across  $\omega^2 \in \{0, \dots, 25\}$  — in both cases the instrument’s insensitivity *is* the content, and in both cases I read through it in the wrong direction.

### 4 Consequences for the article’s claims

1. **The ceiling statement is the robust claim and should carry the headline.** “No panel size buys you more than about two judges” is  $1/\bar{\varphi} = 2.06$ : it names no  $J$ , so the  $J = 32$  correction cannot touch it. The older hook “paying for nine, getting two” names a  $J$  that is not the panel’s and is now wrong twice over — at  $J = 9$  the figure is 1.84, not 2. **Retitle.**
2. **ESDOF should be computed on errors or dropped, and not called  $n_{\text{eff}}$ .** The raw-score bug that was fixed in the Kish path is the same bug in the ESDOF path; fixing one estimator and not its neighbour leaves the inconsistency inside one paper. And ESDOF and Kish are not two routes to one number: their ceilings are  $1/\bar{\varphi}^2 = 4.23$  and  $1/\bar{\varphi} = 2.06$ .
3. **“The item is the unit of replication” stays as Lyra graded it: speculative, not proved** — it is a direction I handed over, and  $\omega^2$  outcome (b) supports it with a remainder rather than establishing it. My own registry node `judge-panel-neff` root stays `in-progress` with the premise `speculative`.
4. **My Proposition 4 stays withdrawn.** Per-item MSE and the error part of  $\text{Var}(M)$  share one ceiling  $1/\bar{\varphi}$ ; the “factor of four” framing was wrong, and conditioning on the item removed  $u_i$ , which *is* the error.